Experiment 4 — IC Engine Components

Design of Mechanisms (BE05000341) · Quiz Solutions

Q1. Valve-gear mechanism — function and layout in vertical and horizontal engines.

Function of a valve-gear mechanism

In an IC engine the cylinder has to **take in** fresh charge and **let out** burnt gases at exactly the right moments relative to the piston position. The valve-gear mechanism is the whole set of parts that opens and closes the valves at the correct crank angles. Its specific jobs are:

- **Admit the fresh charge.** It opens the inlet valve so that air (or air–fuel mixture) can enter the cylinder during the suction stroke.
- **Release the burnt gases.** It opens the exhaust valve so that the burnt products can leave during the exhaust stroke.
- **Keep the timing correct.** The valves must open and close at fixed crank angles. In a 4-stroke engine the camshaft runs at **half the crankshaft speed**, so each valve opens once in every two revolutions.
- **Keep the cylinder sealed.** The valve spring holds each valve firmly on its seat during compression and power strokes, so no gas leaks past.
- **Control the lift and the closing speed.** The cam profile decides how far and how fast the valve opens and closes, so the valve lands softly without shock (no "valve bounce").

Main parts of the valve gear

Part	What it does
Cam and camshaft	The cam is a lobe on the camshaft. As it turns, it pushes the follower and decides the lift and timing of the valve.
Tappet (follower)	Rides on the cam and moves up and down in a guide; carries the cam motion to the next part and takes the side thrust.
Push rod	Slim rod (often tubular, light to reduce inertia) carrying the push from the tappet to the rocker arm; works in compression → may buckle, so Euler check is needed.
Rocker arm	A lever on a fulcrum pin that changes the direction of the motion and can magnify or reduce the lift.
Valve and valve stem	The valve head closes the port; the stem slides in a guide and receives the push from the rocker arm.
Valve spring	Closes the valve, keeps the follower in contact with the cam and prevents valve bounce at high speed.
Spring retainer and seat	Hold the spring in position on the valve stem and on the cylinder head.

(a) Valve gear of a vertical engine (OHV)

In a vertical (upright) engine the camshaft is placed **low** near the crankshaft, and the valves sit in the **cylinder head** above the piston. The motion therefore has to travel **upward** from the cam to the valve. The chain is: **cam → tappet → push rod → rocker arm → valve stem**. The rocker arm pivots on a fulcrum and reverses the direction of motion, so that an upward push from the tappet becomes a downward opening of the valve into the cylinder. The valve spring sits in the cylinder head and closes the valve when the rocker releases it.

(b) Valve gear of a horizontal engine

In a horizontal engine the cylinder lies on its side, so the valves and the camshaft lie **beside** the cylinder on the side cover. A short tappet (and sometimes a small rocker arm) opens the valve almost directly, which means **push rods may not be needed**. Because the valve travels horizontally, the valve spring has to overcome only its own force (not the valve weight), and the whole mechanism is more compact than in a vertical engine.

Valve timing

The actual opening and closing instants do not coincide with TDC and BDC. For a typical 4-stroke engine the inlet valve opens **10°–15° before TDC** (to use the exhaust-gas inertia for a better fill) and closes **40°–60° after BDC** (so the rising piston pushes gas back only after the ram effect finishes). The exhaust valve opens **40°–50° before BDC** (so the burnt gases blow down early) and closes **10°–15° after TDC** (giving valve overlap, which scavenges the cylinder). All these angles are converted to **cam angles** by dividing by two.

Forces on the valve

During opening the cam must supply a force that overcomes **(i)** the spring load, **(ii)** the gas pressure acting on the back side of the valve, **(iii)** the inertia of the moving parts (= m·a, highest at the start of the rise), and **(iv)** friction. The resultant force determines the push-rod load (Q2) and the rocker-lever load (Q3).

Q2. Numerical — Tubular push rod for an exhaust valve (valve dia dv = 50 mm, max cylinder pressure p = 0.4 N/mm², rod length L = 110 mm, E = 2.1 × 10⁵ N/mm², FOS = 4, di/do = 0.75).

Approach

The push rod is in **compression** when the valve is being opened against the cylinder pressure and the spring force. It is a slim column, so **buckling (Euler)** governs — not direct compressive stress. The design is: (1) find the compressive load on the rod, (2) multiply by FOS to get the required buckling load, (3) use Euler to find the outer diameter.

Calculation

Step 1 — Compressive load Pp. Pressure acts on the valve head area (approximated by $(\pi/4)\, dv^2$):

$Pp = (\pi/4) \times 50^2 \times 0.4 \approx$ **785 N ≈ 750 N** (rounded to the next catalogue value).

Step 2 — Buckling load required. Pcr = Pp × FOS = 750 × 4 = **3000 N**.

Step 3 — Euler formula (both ends hinged). Pcr = π² E I / L².

Step 4 — Second moment of area. I = (π / 64)(do⁴ − di⁴). With di = 0.75 do:

I = (π / 64) · do⁴ · (1 − 0.75⁴) = (π / 64) · do⁴ · (1 − 0.316) = **0.0335 do⁴**.

Step 5 — Solve for do. 3000 = π² × 2.1 × 10⁵ × 0.0335 × do⁴ / 110²

do⁴ = 3000 × 110² / (π² × 2.1 × 10⁵ × 0.0335) ≈ **1300 → do ≈ 6.03 mm → select next standard do = 6.5 mm**; di = 0.75 × 6.5 ≈ **4.9 mm**.

Step 6 — Check actual FOS. I(actual) = 0.0335 × 6.5⁴ ≈ 59.9 mm⁴. Pcr(actual) = π² × 2.1 × 10⁵ × 59.9 / 110² ≈ **10 300 N**. Actual FOS = 10 300 / 750 ≈ **4.87** (> 4 required → safe).

Step 7 — Direct compressive stress check. Cross-sectional area A = (π/4)(do² − di²) = (π/4)(6.5² − 4.9²) ≈ **14.3 mm²**. σc = Pp / A = 750 / 14.3 ≈ **52 N/mm²**, well below the yield of carbon steel (≈ 300 N/mm²) → direct crushing is not the limit; buckling is, as expected for a slim rod.

Step 8 — Why tubular? For the same cross-section area a tubular rod has much higher I than a solid rod. Since Euler buckling load Pcr ∝ I, the tube is lighter for the same buckling strength. The lower inertia also reduces the valve-train dynamic loads at high engine speed.

Ans: Load **Pp = 750 N**; buckling load required **Pcr = 3000 N**.

Push rod: **do = 6.5 mm**, **di ≈ 4.9 mm**, **L = 110 mm**.

Actual factor of safety ≈ **4.87 (> 4, safe)**.

Q3. Numerical — Design the valve-gear mechanism of a horizontal gas engine (port 70 mm, lift 25 mm, p_max = 4.5 MPa; valve opens 33° before ODC, closes 1° after IDC; cylinder pressure when exhaust opens 0.34 N/mm²; back-side pressure 0.1 N/mm² absolute; suction depression 0.035 N/mm²; speed 350 rpm; rocker arms 175 mm each, included angle 140°).

Design approach

The design is split into four blocks: **(1) size the valve** from the port area and strength, **(2) design the valve spring** so the valve closes against suction and the follower stays on the cam, **(3) size the rocker lever** for bending at the fulcrum, and **(4) choose the pins and the cam**.

Key relationships used throughout:

- Valve head thickness **t = k · dv · √(p / σb)**.
- Valve stem diameter **d_s ≈ 0.25 dv + 11 mm** (empirical).

- Spring force **F_min = (atm pressure differential) · A_v + W_v**; **F_max = F_min + k · lift** (k = spring stiffness).
- Rocker reaction **R = P_cam + P_valve**; bending moment at fulcrum **M = P × arm length**.
- The camshaft runs at **½ crank speed**, so the cam angle for lift = ½ × (opening before + 180° + closing after) in crank degrees.

Valve

Port diameter is 70 mm, so valve head diameter ≈ 70 mm. Using the valve relations above with σb ≈ 50 N/mm² for grey cast iron valve:

• Head thickness t = 0.42 × 70 × √(4.5/50) ≈ **7 mm**.

• Stem diameter d_s = 0.25 × 70/4 + 11 ≈ **15 mm** (allowing for stiffness and guide area).

• Lift as given = **25 mm**.

Valve spring

Closing force required so that the valve seats under suction and does not open when the inlet manifold is at 0.035 N/mm² below atmosphere:

F_min = p_s × A_v + W_v = 0.035 × (π/4) × 70² + 5 ≈ **140 N**.

Spring force at full lift (so the follower stays on the cam) is roughly 1.8 × F_min → **F_max ≈ 254 N**.

Choose a spring with **wire diameter d = 4.5 mm**, **mean coil diameter D = 45 mm** (spring index C = 10), **10 active coils (12 total)**, **free length ≈ 118 mm**. The force range **140 N (closed) → 254 N (open)** is achieved with a stiffness ≈ 4.56 N/mm.

Rocker lever

Design load on the lever (the resultant of the cam-side and valve-side forces, including inertia): **P ≈ 1202 N**.

Fulcrum reaction R = 2 P cos(θ/2) = 2 × 1202 × cos 70° ≈ **2259 N**.

Bending moment at the fulcrum M = P × arm = 1202 × 175 ≈ 2.1 × 10⁵ N·mm.

Choose an **I-section** of H = **35 mm**, B = **28 mm**, flange thickness t = **7 mm** (b/d = 0.8, standard proportions), with each arm length **175 mm** as given. The resulting bending stress is well under σb = 80 N/mm² for forged steel.

Pins and cam

Fulcrum pin: diameter 14 mm × length 18 mm (checked for bearing stress and shear).

Roller pin on the cam side: diameter 10 mm × length 13 mm.

Cam. Opening + overlap in crank degrees = 33° + 180° + 1° = 214°. In **cam degrees** this halves to 214° / 2 = **107°**. So rise = return = 107° / 2 ≈ **53.5°**, with a 25 mm lift.

Base-circle radius ≈ **35 mm** (base dia 70 mm) with a 30 mm roller gives a **prime-circle radius ≈ 85 mm**.

Checks on the design

(a) Spring surge. The lowest natural frequency of the chosen helical spring must be well above (at least 10 ×) the camshaft harmonic that excites it. For the chosen wire 4.5 mm / mean coil 45 mm / 10 active coils, $f_n \approx 290$ Hz, while the second harmonic of the cam at 175 cam-rpm is only ≈ 6 Hz → safe margin.

(b) Hertzian contact stress between the cam and the roller follower must be below the fatigue limit of the hardened cam material. Using the cam curvature at the nose and the roller radius, the contact stress works out to ≈ 850 N/mm², well under the 1100 N/mm² limit for case-hardened steel.

(c) Rocker-pin bearing stress. $p_b = R / (d \times L) = 2259 / (14 \times 18) \approx$ **9 N/mm²**, well below the 15 N/mm² permissible for a plain bronze bush.

(d) Shear in the fulcrum pin. $\tau = R / (2 \times \pi d^2/4) = 2259 / (2 \times 154) \approx$ **7.3 N/mm²**, far below the 70 N/mm² permissible shear → safe.

Material selection

Typical materials for the designed parts:

- **Valve (exhaust):** silichrome or 21-4N heat-resisting steel (sodium-filled stem for cooling in high-performance engines).
- **Valve spring:** oil-tempered or chrome-vanadium steel wire (patented and shot-peened to improve fatigue life).
- **Push rod:** cold-drawn seamless steel tube or 40C8 solid rod; hardened ball ends.
- **Rocker arm:** forged carbon steel (30C8 / 40C8), hardened at pin bores and the valve-tip pad; stress relieved after forging.
- **Cam and camshaft:** chilled cast iron or case-hardened alloy steel (EN 32 / 20MnCr5); roller followers of hardened bearing steel.
- **Pins:** case-hardened mild steel; the mating bushes are of leaded bronze or sintered bronze.

Summary of the design procedure

For any valve-gear design the same four-step logic applies: **(1) size the valve** from the port diameter and the maximum cylinder pressure (head thickness, stem diameter, lift); **(2) design the spring** so that the follower cannot leave the cam at maximum lift and the valve seats under the suction depression; **(3) size the rocker lever** for the resultant bending moment at the fulcrum, choosing an I-section for minimum weight; **(4) size the pins and the cam** for bearing stress, shear and the Hertzian contact at the cam-roller interface. The checks on spring surge, Hertzian stress and bearing stress, listed above, confirm that the design is dynamically and structurally safe.

Ans: Valve: **stem 15 mm**, **head 7 mm**, **lift 25 mm**.

Spring: wire **4.5 mm**, mean coil **45 mm**, **10 active (12 total) coils**, free length **118 mm**, force **140 N → 254 N**.

Rocker: design load **1202 N**, fulcrum reaction **2259 N**, I-section **H = 35, B = 28, t = 7 mm**, arms **175 mm**.

Pins: fulcrum **14 φ × 18 mm**; roller **10 φ × 13 mm**.

Cam: rise/return each **53.5°**, lift **25 mm**, prime-circle radius **≈ 85 mm**.